



Scroll. It's all *real*.¹

Ayush Yadav — software engineer · Cincinnati, Ohio

the capstone — agentic automl: seven gated phases —

case files: Applied · Glyph · Cadence

scroll — the story starts here

in a hurry — the work · the evidence · the resume

1. every claim on this page links to its evidence. ↩

01 / 07

This is a record of what I've built. *And how each piece was checked before I called it done.*

I'm a recent computer-science graduate from Miami University. I build the whole path — the interface, the services behind it, the data they run on, and the machine learning when the problem calls for it. The claims on this paper carry receipts.

ap·prov·al

/ ə'pru:vəl / · noun

1. the moment a human signs for a machine's work. 2. the step I leave to a person.

n.b. — every project below opens into its own case file, source, or live build.

¶ 03 / 07 · the path — field records · 12:06

Thousands of service tickets. Zero structure.

*This is where I learned that the work
starts in the mess.*

jun 2025 — may 2026
miami university · oxford, ohio

ITSM Data Integration Student Associate

jun 2025 —

The tickets arrive without shape. A million-plus service records, and no two name their problem alike.

1m+ operational records — high-volume python + sql transforms

the fix —

Python and SQL teach the records shape. The transforms feed OAS and Tableau, and the dashboards they fill are read across university departments.

oas/tableau reporting — dashboards across departments

the bot —

A chatbot in Slack takes the routine questions — wired to Workday, Oracle Analytics, and Tableau, with access governed at every joint.

governed data access — workday · oracle · tableau apis

the assistant —

A second assistant reads policy and answers with its sources cited — and drops the answers whose quotes it cannot verify.

19/20 cited-source sweep — policybot, the policy assistant

The agent drafts the whole pipeline.

Nothing runs until a human says go.

Agentic AutoML turns a dataset and a goal into a structured, auditable ML workflow. The agents plan it and run it, then stop at the seventh phase and wait for a person.

7 phases · every one gated · langgraph + mcp orchestration

what the run is held to — quoted from the case file

Require human approval before generated actions alter workflow state.

read the case file — agentic automl platform —

```
1.0 ingest
2.0 explore
3.0 preprocess
4.0 engineer
5.0 train
6.0 evaluate
    the human gate - go / no-go
7.0 deploy
```

fig. 4.0 - the 7-phase gated lifecycle, abridged.

run · model	metrics	status
038 · log-reg		approved

fig. 4.1 - experiment registry, transcribed private-safe excerpt.
 - not published: the repository is private, and the demo-data metric trail has not shipped. see the case file.

Applied

Your inbox already knows where you applied.
Applied reads it - and the classifier runs in your browser.

macro-f1 0.98 - 96-sample gate
the case file —
last verified 2026-07

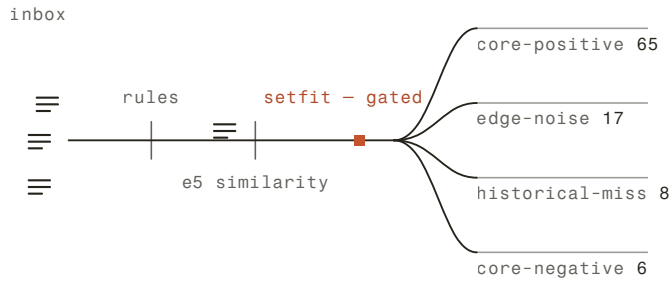


fig. 5.0 – the sorting line – the 96-message eval set through rules → e5 similarity → gated setfit.
96 messages · 12 per label × 8 labels · macro-f1 0.9791 – committed eval baseline · ships in-browser: int8 onnx, 22.8 mb

Glyph

No frameworks. Just math, memory, and C++.

Then OpenMP and SIMD together made it 3.5× faster.

openmp+simd dot kernel – 3.5x vs
-O3 baseline, committed
benchmarks

the case file —

last verified 2026-07

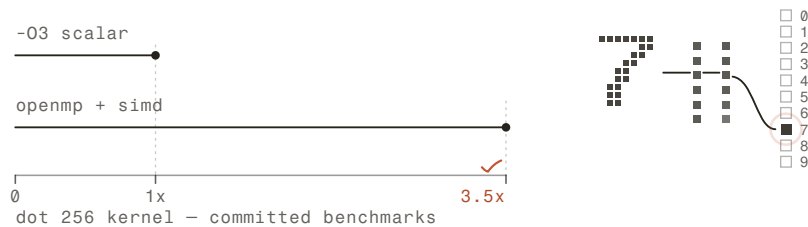


fig. 5.1 – the race – equal time, measured distance · and the forward pass, drawn.
3.5x – openmp+simd dot kernel vs -O3 baseline (dot 256), committed benchmarks · avx-512 · avx2 · neon · wasm-simd128

jetpack-compress (<https://jetpack-compress.vercel.app>)

One gzip stream. Every core writing it.

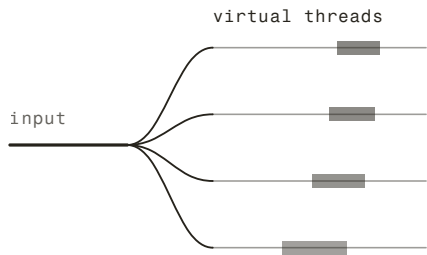
*jetpack-compress splits the bytes across virtual threads
– and stitches one byte-valid member.*

72 tests, 0 failures – jdk 25
@ af2c4b1 (<https://github.com/yadava5/jetpack-compress/blob/af2c4b1/README.md>)

the live demo ↗ (<https://jetpack-compress.vercel.app>)

source (<https://github.com/yadava5/jetpack-compress>)

last verified 2026-07



gzip this figure in your browser [→](#)

fig. 5.2 – split → parallel → stitch – virtual-thread gzip into one byte-valid member.
~6.5x vs single-threaded java.util.zip (±50%, quick benchmark)
· 72 tests, 0 failures · jdk 25 @ af2c4b1

Cadence

A calendar and task app you drive in plain English.

Type a sentence and it files the event or task – and now schedules Google Meet meetings with multi-attendee invites.

1,145 automated tests – 634 frontend + 511 backend, vitest
the case file [→](#)
last verified 2026-07

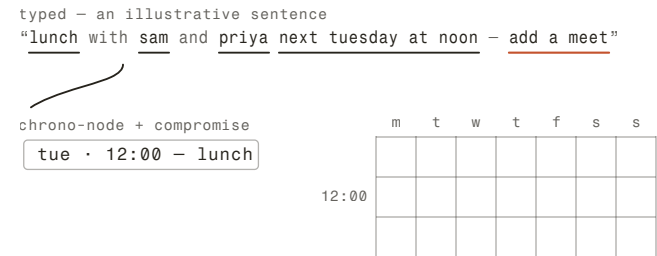


fig. 5.3 – the parse – a plain-english sentence resolves into chips on the week grid + a google meet link.
chrono-node + compromise · google meet – multi-attendee invites · 1,145 tests passing – taskflow-calendar @ 69a59e7

also on file –
master inventory pipeline policybot

Make it learn.

182 backend tests — jobtracker validation ledger

Make it fast.

openmp+simd dot kernel — 3.5x over -O3, committed benchmarks

Make it honest.

19/20 cited-source sweep — policybot validation ledger

fig. 6.0 — the litany, with receipts.

gate	disposition
classifier gate — jobtracker	✓ passed
cited-source sweep — policybot	— refused
benchmark suite — fast-mnist	— refused

fig. 6.1 — the gates, kept.
refused — the gate stopped the run ·
passed — human signed off

Every pipeline I build ends with a
human decision.

This one ends with yours.

Ayush Yadav

availability — open to new-grad software, data, and ml engineering roles
cincinnati, ohio · it's 3:50 am here right now

aesh_1055@icloud.com resume github (<https://github.com/yadava5>)
linkedin (<https://www.linkedin.com/in/ayush-yadav-developer/>)

Email me — I read everything.



references — footnote 1, kept.

1. macro-f1 0.9791, 96-sample gate	jobtracker — receipt 05 —
2. 3.5x — openmp+simd dot kernel vs -O3 baseline	fast-mnist-nn — receipt 02 —
3. 1,145 automated tests — 634 frontend + 511 backend	taskflow-calendar — receipt 01 —
4. 72 tests, 0 failures on jdk 25 jetpack-compress — readme @ af2c4b1 ↗ (https://github.com/yadava5/jetpack-compress/blob/af2c4b1/README.md)	
5. 19/20 cited-source sweep	policybot — receipt 01 —
6. 1M+ operational records transformed	master-inventory — receipt 05 —
7. 7-phase gated lifecycle, human go/no-go	automl — the case file —